ANDISA KHUMALO

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PROFESSIONAL PROFILE

Final-year BEng Electrical and Computer Engineering student (68% average), ranked third-best in the final year for Semester 1 2026, and a South32 bursar seeking a Graduate Engineer role in heavy industry. Hands-on PLC experience with Delta and Siemens controllers and CODESYS, including a pharmaceutical clean-room control system with fail-safe E-stop, interlocks and latched alarm handling. Also built a CNN-driven autonomous vehicle, presented at the international SPETM conference, and uses Python and MATLAB for system analysis. Based in Richards Bay and available from December 2026.

EDUCATION

University of Zululand *2023 – 2026*

Bachelor of Engineering (Electrical and Computer Engineering), completing November 2026

- **Average:** 68%. Certificate of Achievement as Third-Best Final-Year Electrical and Computer Engineering student, Semester 1 2026 (awarded September 2026).
- **Relevant modules:** Process Control & Instrumentation, Control Systems, Power Systems, Engineering Systems Design, Communications & Networks, Occupational Health & Safety

Qhamuka High School *2021*

National Senior Certificate (Matric)

ENGINEERING PROJECTS

Pharmaceutical Clean-Room Control System (PLC, individual project) *2026*

Process Control & Instrumentation, based on an Aspen Pharmacare (Gqeberha) case study

- Designed a 19-rung IEC 61131-3 ladder program handling 9 digital inputs and 6 outputs for clean-room HVAC, UV sterilisation and production control.
- Implemented a timed start-up sequence (supply fan, return fan after 5 s, 6 s pressure stabilisation, 7 s UV warm-up) before the room is qualified for production.
- Built fail-safe protection: a normally-closed E-stop that de-energises all outputs, a door lock tied to UV status, production blocked while any alarm is active, and latched faults reset only by HMI acknowledgement.
- Programmed and tested in CODESYS with an HMI visualisation, then wired the physical I/O on a Delta PLC; all 8 fault and operating test scenarios passed, with the room qualifying in about 18 s.

Autonomous Electric Vehicle (Final-Year Capstone) *2026*

- Designed, built and tested a low-cost autonomous RC vehicle on a Raspberry Pi 4B, integrating camera perception, embedded control and motor actuation.
- Trained a convolutional neural network (CNN) for lane-following steering on 1,620 labelled image–steering pairs (steering-angle MAE 16.66°, RMSE 25.03°).
- Completed 10 laps of the test track autonomously with no human intervention, steering itself back into the lane whenever it drifted out.
- Integrated an OV5647 camera, PCA9685 PWM controller, steering servo, BTS7960 motor driver and ultrasonic sensors for autonomous steering and experimental obstacle detection.
- Built an Android app (Kotlin, Jetpack Compose) with a Flask backend on the Pi for live video, manual control and autonomous-mode start-up.

PLC Practicals: Siemens and Delta *2026*

- **Warehouse forklift safety system (Siemens PLC):** combined a digital capacitive entry sensor with an analogue ultrasonic sensor scaled to millimetres, using comparators to sound a buzzer in a 60–200 mm danger zone, with latched warnings and push-button acknowledgement.
- **Home burglar alarm (Delta PLC, WPLSoft/ISPSoft):** ladder logic with a normally-closed foil sensor, motion detector and 60 s arming-delay timer, with indicator lights for each sensor and the alarm.

AK Mechanics: Vehicle Fleet Management Web App (Graduate Attribute Project) *2026*

- Built a full-stack web application in ASP.NET Core MVC with Entity Framework Core and SQL Server, with role-based access for clients, service providers and administrators (ASP.NET Identity).
- Deployed on Microsoft Azure App Service with an Azure SQL Database.

Smart Airport Luggage Tracking System (Group Capstone, design proposal) *2026*

- Designed the tracking architecture combining RFID identification, GPS positioning and IoT sensing, with LoRa and MQTT communication to a central database and web/app interface.
- Developed the tracking algorithm and sensor subsystem, and contributed to the system proposal and execution plan.

CERTIFICATIONS & AWARDS

South32 Bursary: awarded on academic merit.

SMR Machinery and Equipment (Distinction), National Research Nuclear University MEPhI (Russia) and University of Zululand, October 2026.

openEuler Linux System Administration, Huawei ICT Academy, June 2026.

openGauss Database Administration, Huawei ICT Academy, June 2026.

TECHNICAL SKILLS

Industrial Automation: Delta and Siemens PLC programming, CODESYS (incl. HMI visualisation), ladder logic (IEC 61131-3), analogue signal scaling, safety interlocks, PID control, instrumentation, SCADA concepts

Control & Simulation: MATLAB/Simulink, PID design, system modelling and simulation

Programming: Python, C/C++, C#, MATLAB, Kotlin (Android), Arduino

Embedded Systems: Raspberry Pi, Arduino, STM32, serial communication, sensor and camera integration, motor control

Computer Vision & AI: OpenCV, CNN training and deployment, lane keeping, dataset collection

Software & Web: ASP.NET Core MVC, Entity Framework Core, SQL Server, Azure, HTML/CSS, Git/GitHub

Documentation: IEEE-format technical reports, system architecture documentation

PROFESSIONAL ACTIVITIES

1st Sustainable Power, Energy, Transportation and Materials (SPETM) Conference: Poster Presenter *2026*

- Presented a research poster at the University of Johannesburg on the design of an electric autonomous vehicle to international researchers, explaining its system architecture, control strategy and machine-learning workflow.

Eskom Expo for Young Scientists, Richards Bay: Judge *August 2026*

- Evaluated about 10 engineering, electronics, robotics, coding and science projects against the programme criteria, questioned participants and contributed to grading and selection.

South African Institute of Electrical Engineers (SAIEE): Student Member *2024 – Present*

- Attend technical talks and industry workshops, and network with practising engineers.

Industrial Site Visit: Scorpio Substation, Richards Bay

- Toured the substation next to the South32 Hillside Aluminium smelter.

ADDITIONAL INFORMATION

Driver's Licence: Code 10

Availability: Richards Bay; available to start from December 2026

REFERENCES

Dr O.E. Oni: Lecturer & Supervisor, University of Zululand | onio@unizulu.ac.za

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